

SEO Plan

Nano Computing ICT Solutions

SITE	nanocomputingict.com
PREPARED	2026-10-03
STACK	Next.js 16.3.6 (App Router, Turbopack) · React 19 · Tailwind v4
SCOPE	all 5 public routes, with the services page treated in depth

0. How to read this document

Sections are split into three tiers:

TIER	MEANING
● BUG	Something actively broken right now, costing rankings today
● GAP	Missing but not yet costing traffic
● OPPORTUNITY	New surface area or capability to build

Every claim in sections 1–3 was **verified against the running site or a live API response** during preparation. Data-collection method and limitations are stated in section 9.

1. Executive summary

The site scores **100/100 on Lighthouse SEO** and is, by that measure, perfect. It is not. Lighthouse's SEO category runs **11 audits**, and the audit responsible for the single most damaging bug on this site passes cleanly for the wrong reason.

The three findings that matter most:

1. ● **Every page canonicalises to the homepage.** `/services`, `/contacts`, `/about` and `/shop` all emit `<link rel="canonical" href="https://nanocomputingict.com">`. This instructs Google to treat four pages as duplicates of the home page and drop them from the index. It is set in one place — `app/layout.tsx:68-69` — and it is almost certainly why these pages do not rank for anything.
2. ● **The home page `<h1>` is not a heading, it is a rotating tagline.** It reads "Your Integrated Safety Partner", where the second half is a `FlipTitle` that mutates every 3 seconds between *Safety Partner*, *ICT Partner* and *Security Partner*. The primary heading on the most important page is non-deterministic, and contains no service or location keywords.
3. ● **There is no structured data anywhere.** Zero JSON-LD across all routes. For a local business this is the largest single missed opportunity: no rich results, no knowledge-panel eligibility, and no structured entity for AI/LLM surfaces to cite.

Underneath those: no sitemap, no robots file, no OG image (despite declaring `twitter:card = summary_large_image`), no `metadataBase`, and two near-empty pages carrying almost no content.

None of this requires a content marketing budget to begin fixing. The overwhelming majority of the work in this plan is engineering, not writing.

2. Current state audit

Measured against the dev server, 2026-10-03.

2.1 Route-by-route

ROUTE	TITLE	DESC LEN	CANONICAL	H1	H2	JSON-LD	OG:IMAGE
/	Your Integrated Safety Partner	295	homepage ●	rotating tagline ●	1	0 ●	missing ●
/services	Services nano computing...	268	homepage ●	Our Services	5	0 ●	missing ●
/contacts	Contact us nano computing...	218	homepage ●	Contact us	0	0 ●	missing ●
/about	About nano computing...	85	homepage ●	About	0	0 ●	missing ●
/shop	Shop nano computing...	88	homepage ●	Shop	0	0 ●	missing ●

All images have `alt` attributes (3 per route, zero missing) `lang="en"` is set correctly Heading order is valid on every page (single h1, h2s nested below)

2.2 ● Missing infrastructure

FILE	STATUS	CONSEQUENCE
app/sitemap.ts	absent	No XML sitemap. Google discovers pages only via internal links.
app/robots.ts	absent	No crawl directives, no sitemap pointer, no AI-crawler policy.
app/opengraph-image.*	absent	No social share image generated.
app/not-found.tsx	absent	Generic 404. Opportunity for helpful recovery links.
app/manifest.ts	absent	No web app manifest.

2.3 Why Lighthouse says 100

The SEO category contains 11 audits. Reading the raw report:

```
canonical    score=1 ← passes: URL is a valid absolute URL.
              It does NOT check that the canonical is
              self-referencing or unique per page.
structured-data  score=null ← "not applicable" because nothing is there.
              Absence is not scored as failure.
robots-txt     score=null ← manual/unaudited.
is-crawlable   score=1
sitemap       NOT IN CATEGORY AT ALL
```

structured-data and robots-txt return null, not 0. **Missing is scored as "nothing to check", not "failed"**. That is the entire gap between the score and reality, and it is why this plan exists: a green Lighthouse report is not evidence of an indexable site.

3. ● Bug: canonical URL collapse

The defect

app/layout.tsx:68-70 :

```
alternates: {
  canonical: "https://nanocomputingict.com",
},
```

Metadata is inherited down the tree. Declaring a canonical once in the **root** layout stamps that same URL onto **every** descendant route. Verified output:

```
/      → <link rel="canonical" href="https://nanocomputingict.com">
/services → <link rel="canonical" href="https://nanocomputingict.com">
/contacts → <link rel="canonical" href="https://nanocomputingict.com">
/about    → <link rel="canonical" href="https://nanocomputingict.com">
/shop     → <link rel="canonical" href="https://nanocomputingict.com">
```

Why it is fatal

A canonical is a *consolidation hint*: "index this URL instead." Applied to `/services`, it tells Google the services page does not deserve its own index entry. Consequences:

- Only the homepage can rank. `/services` cannot outrank `/` for anything.
- Signals that would build authority on `/services` (links, engagement, quotes) are consolidated onto `/`, discarding them.
- Any future service page inherits the same defect, so the site structurally cannot grow a long-tail footprint.

This is a hard blocker. Nothing else in this plan produces measurable gains until it is fixed.

The fix

Remove `alternates` from `app/layout.tsx`. In App Router, a route's own metadata resolves to its own URL automatically once a root canonical no longer overrides it.

Verify each route then emits a self-referencing canonical:

```
// app/services/page.tsx
export const metadata: Metadata = {
  // ...
};
// No `alternates` needed — Next emits https://<host>/services
```

If a canonical must be explicit, set it per route, never in the root layout:

```
alternates: { canonical: "/services" } // relative — resolved against metadataBase
```

Related: `metadataBase` is missing

`app/layout.tsx` never sets `metadataBase`. Next.js therefore cannot resolve relative OG/Twitter/canonical URLs to absolute ones and falls back to `http://localhost:3000` during development. Once real OG images are added (§7), this must be set first:

```
metadataBase: new URL("https://nanocomputingict.com"),
```

4. ● Bug: the home page h1 is a rotating tagline

`components/home-hero.tsx:35-44` :

```
<h1 id="hero-title" ...>
  Your Integrated{" "}
  <FlipTitle
    words=[["Safety Partner", "ICT Partner", "Security Partner"]]
    className="px-0 text-white"
  />
</h1>
```

`FlipTitle` → `FlipWords` re-renders the second half on a 3-second interval (`components/flip-title.tsx:14` , `duration = 3000`).

Two independent problems:

a. It is not a keyword-bearing heading. "Your Integrated Safety Partner" is a brand tagline. The home page's single `<h1>` should state what the business is and where it operates. Neither is present.

b. The h1 mutates. The served HTML and the rendered text differ, and the rendered text changes every 3 seconds. This is exactly the kind of unstable, non-deterministic primary heading that undermines both crawling and AI/LLM extraction, which typically sample a single render.

Fix. Make the h1 static and descriptive; keep the rotation as a decorative element below it.

```
<h1 id="hero-title" className="...">
  CCTV, access control and ICT infrastructure
  <span className="block text-white/60">in Addis Ababa, Ethiopia</span>
</h1>
<p className="...">
  Your integrated safety partner — CCTV and access control, time
  attendance, networks, servers and software, designed, installed and
  maintained by our engineering team.
</p>
```

`FlipTitle` remains valid inside `<p>` (it is already used this way in `home-hero.tsx:46` 's sibling and in `contact-section.tsx:61`). Do not put it in a heading again.

5. ● Thin content on /about and /shop

Both pages now render a title and nothing else (85 and 88 character descriptions). Thin pages are a liability:

- They generate no long-tail surface area.
- They can be classified as **soft 404s** or low-value pages, which drags crawl budget perception.
- They are the weakest possible destination for any internal link.

Options, in order of preference:

1. **Build them out.** /about → company story, team, the standards work is held to, the 7+ years / 200+ projects claims already in `app/layout.tsx` . /shop → the actual product categories (Network Devices, Computer Accessories, Computers — these exist as real categories on the live WordPress site and in `site-footer.tsx`).
2. **Merge / noindex** if neither will ever get real content.

Leaving them as title-only stubs is not a viable steady state.

6. Content and keyword strategy

6.1 Method and its limits

Data was gathered via **986 live calls** to Google Suggest, with recursive 2-level expansion — **4,743 unique suggestion strings**, 4,210 after noise filtering.

No volume data exists in this plan. Google Suggest returns no search volumes, CPCs or difficulty figures. Suggest frequency is a proxy for demand, not a measurement of it. Any real volume number would require Keyword Planner, Ahrefs, Semrush or DataForSEO. Relative demand here is inferred from **how consistently a term surfaces across independent seed queries and geo variants** — a defensible but imperfect signal.

6.2 ● The headline opportunity: empty local queries

These seeds returned **zero suggestions**, re-confirmed individually:

```
cctv maintenance addis ababa    → 0
cctv repair addis ababa         → 0
server installation addis ababa  → 0
lan cabling addis ababa         → 0
access control system addis ababa → 0
access control system price in ethiopia → 0
firewall installation addis ababa → 0
server installation ethiopia     → 0
```

Google discards the geo modifier and falls back to generic completions. The standard reading: **nobody — including every competitor — has built search demand into these terms.** There is no established SERP to displace, and no competitor holding the position.

Equally, `access control system price` surfaces geo variants for **Bangladesh, Kenya, Nigeria, Pakistan, Ghana, Nepal, Uganda, India — Ethiopia is absent.** Competing countries own this space.

This is a real opening, and it is cheap: these are pages a competitor would have to build from scratch.

6.3 Verified geo-flagged keywords

Google's own `firefox`-client suggest responses carry `google:suggestsubtypes`, where codes `22 / 30` mark **Google's own geo-modification of the query**. This is a stronger local-intent signal than string matching. **149 suggestions carried the flag; 137 are service-relevant**. Highest-value subset, flagged under both `gl=et` and `gl=us` (locale-stable):

KEYWORD	SERVICE
security camera installation in addis ababa	CCTV
cctv installer in addis ababa	CCTV
security camera installation in ethiopia	CCTV
cctv camera installation in ethiopia	CCTV
cctv camera installation price in ethiopia	CCTV
cctv camera price in addis ababa	CCTV
security camera price in addis ababa	CCTV
cctv camera shop in addis ababa	CCTV / retail
security camera supplier in addis ababa	CCTV / retail
cctv camera supplier in ethiopia	CCTV / retail
cctv camera importer in ethiopia	CCTV / retail
computer repair in addis ababa	Repair
laptop repair addis ababa	Repair
macbook repair addis ababa	Repair / Apple
computer maintenance in addis ababa	Repair
data recovery addis ababa	Repair
networking company in addis ababa	Networking
web development company in ethiopia	Web
website design company in ethiopia	Web
web developer in addis ababa	Web

`gl=et` -only, also verified: time attendance machine in ethiopia · biometric attendance machine price in ethiopia · attendance machine price in ethiopia · surveillance camera price in ethiopia · outdoor cctv camera price in ethiopia · wifi service in ethiopia · wifi cost in ethiopia · firewall price in ethiopia · cisco firewall price in ethiopia · access switch price in ethiopia · network cable price in ethiopia · network trunk price in ethiopia · door lock price in ethiopia · website development price in ethiopia · website development cost in ethiopia · app developer in ethiopia · mobile app developers in ethiopia · security service providers in ethiopia · security systems in ethiopia · home security system in ethiopia · cctv installation training · cctv and alarm installation courses

Note the **pricing cluster**: price in ethiopia / cost in ethiopia recurs across CCTV, attendance, networking, web and doors. See §6.5.

6.4 Competitor landscape

COMPETITOR	DOMAIN	POSITIONING OBSERVED
Safe Technology PLC	safetechet.com	Broadest service list — CCTV, security fence, gate automation, access control, attendance, parking barrier, bollard, video intercom, fire alarm, PBX, PA, data networking. Strong brand recall in Suggest.
SOLFET	solfet.com	Authorized Dahua distributor. Explicit "CCTV Company in Ethiopia", "Access Control Company in Ethiopia". FAQ schema.
Seventech Solutions	seventech.et	Geo-slug pages: /time-attendance-machine-in-ethiopia/ , /structured-network-cabling-in-ethiopia/
Abay Tech Solutions	abaytechsolutions.com	/products/security-camera — "CCTV & Security Camera Installation in Ethiopia"
iSense Technologies	isense-tech.com	Founded 2007. /services.html with integration narrative.
AddisTech	addistechsolutions.com	ZKTeco distributor. Google rating 3.7.
IE Networks	ienetworksolutions.com	"Largest IT Company in Ethiopia" — enterprise positioning.
Prokal Technologies	prokaltech.com	UNV + ZKTeco + BDCOM distributor; Bole/Kality/Akaki geo copy.
Mac & More	mac-more.com	Claims Apple Authorized Service Provider in Ethiopia — outranks Nano on Mac.

Google autocomplete names four of these as recognised local entities — safe tech security camera cctv addis ababa , salem computer maintenance and networking addis ababa , server engineering and trading plc addis ababa , wubsites website design and development service in addis ababa . Nano Computing appears in **none**, despite holding a 5.0 Google Places rating.

6.5 Verified content gaps in the market

1. **No geo-suffixed URLs anywhere in the category.** Every competitor uses a flat root. Nobody owns /cctv-installation-addis-ababa/ .
2. **Almost no published pricing.** Only one outlier (DreamTech: "Packages start at ETB 15,000, including VAT"), while price in ethiopia suggestions are pervasive. Nobody captures this.
3. **Repair and data recovery are the strongest verified demand with the weakest site coverage.** computer repair in addis ababa , laptop repair addis ababa , macbook repair addis ababa , data recovery addis ababa are all geo-flagged — and Nano sells repair services (per the live site) while the current build has no repair page at all.
4. Competitors rank for **noun-phrase brand lookups** (networking company in addis ababa), not "get me a quote" intent. Informational, low-commitment.

6.6 Keyword map

CLUSTER	PRIMARY TARGET PAGE	SUPPORTING TARGETS
CCTV installation	<code>/services</code> (§8) + <code>/services/cctv-installation</code>	price, supplier, shop, importer
CCTV maintenance & repair	new <code>/services/cctv-maintenance</code>	zero-suggestion geo terms — no competition
Access control	<code>/services</code> + new <code>/services/access-control</code>	door lock price, biometric
Time attendance	<code>/services</code> + new <code>/services/time-attendance</code>	attendance machine price in ethiopia
Networking	<code>/services</code> + new <code>/services/networking</code>	lan cabling addis ababa, firewall price
Web & mobile	<code>/services</code>	web development price/cost in ethiopia
Computer repair	new <code>/services/computer-repair</code>	laptop, macbook, data recovery — verified demand, zero coverage
Pricing	new <code>/pricing</code>	the whole price in ethiopia cluster
Brand/entity	<code>/about</code>	7+ years, 200+ projects, 24/7 support

7. Technical infrastructure to build

7.1 app/sitemap.ts ●

```
import type { MetadataRoute } from "next";

export default function sitemap(): MetadataRoute.Sitemap {
  const base = "https://nanocomputingict.com";
  const now = new Date();
  const routes = [ "", "/services", "/contacts", "/about", "/shop" ];
  return [
    ...routes.map((r) => ({
      url: `${base}${r}`,
      lastModified: now,
      changeFrequency: "monthly" as const,
      priority: r === "" ? 1 : r === "/services" ? 0.9 : 0.6,
    })),
    // add each new service page here as it ships
  ];
}
```

Regenerate automatically when service pages are added in §9.

7.2 app/robots.ts ●

```
import type { MetadataRoute } from "next";

export default function robots(): MetadataRoute.Robots {
  return {
    rules: { userAgent: "*", allow: "/" },
    sitemap: "https://nanocomputingict.com/sitemap.xml",
    host: "nanocomputingict.com",
  };
}
```

Note: do **not** disallow `/_next/`. Next.js serves required assets from there; blocking it breaks rendering for every crawler.

7.3 app/opengraph-image.tsx ●

Every route declares `twitter:card = summary_large_image`, which requires an image. None exists — social shares currently render as bare text links.

```
import { ImageResponse } from "next/og";

export const alt = "nano computing ICT Solutions — CCTV, access control and networks in Addis Ababa";
export const size = { width: 1200, height: 630 };
export const contentType = "image/png";

export default async function OG() {
  return new ImageResponse(
    (
      <div style={{ display: "flex", flexDirection: "column", justifyContent: "center",
        width: "100%", height: "100%", background: "#0a0a0a", color: "#fff",
        padding: 80, fontFamily: "sans-serif" }}>
        <div style={{ fontSize: 64, fontWeight: 800 }}>nano computing ICT Solutions</div>
        <div style={{ fontSize: 32, color: "#a1a1aa", marginTop: 16 }}>
          CCTV · Access Control · Networks · Servers · Software
        </div>
        <div style={{ fontSize: 28, color: "#71717a", marginTop: 12 }}>Addis Ababa, Ethiopia</div>
      </div>
    ),
    { ...size }
  );
}
```

Place in `app/` to apply site-wide, or per route to customise. Set `metadataBase` (§3) first.

7.4 app/not-found.tsx ●

A 404 that links to `/services`, `/contacts` and the shop recovers stranded visitors and distributes internal links.

7.5 llms.txt ●

`app/llms.txt/route.ts` serving a plain-text summary of services + contact details. Increasingly read by AI assistants; low cost, no downside, no established ranking effect — treat as cheap experiment, not a ranking lever.

8. The services page (in depth)

This is the site's highest-value asset and the right target for most investment. It currently has 5 well-structured `<h2>` service sections and the strongest on-page copy on the site — a good foundation that is being actively suppressed by the canonical bug.

8.1 Position the page for the primary cluster

`/services` should own **"CCTV installation Addis Ababa"** and the service-category head terms. It cannot rank for them until §3 is fixed.

8.2 Per-service depth, not just anchors

The services page is a single scrolling document. A visitor who needs CCTV pricing should not have to read past four unrelated sections. Recommended IA:

```
/services          overview — all five, short scannable summaries
/services/cctv-installation  full page: coverage types, HD vs IP, NVR, remote view,
                             maintenance, pricing bands, FAQ, embedded video
/services/access-control
/services/time-attendance
/services/networking
/services/computer-repair  ← verified demand, currently zero coverage
```

Each child page: unique `title` and `description`, an H1 with the geo term, 600–1000 words of genuinely useful local content, an FAQ block, and its own JSON-LD (§9). Link each child up to `/services` and back down with `breadcrumbList` markup.

This also **fixes the architecture for long-tail**: one durable page per service beats five sections on one page.

8.3 On-page copy corrections

Current per-service `summary` and `points` (in `app/services/page.tsx`) are strong, but they contain **no location language at all** — zero occurrences of "Addis Ababa" or "Ethiopia" in any of the 5 sections. Natural, non-stuffed insertions:

- `service-section.tsx` subtitle area → "Installed and maintained across Addis Ababa."
- Each section's closing line → service + location, e.g. "CCTV installation in Addis Ababa for shops, offices and homes."
- The page intro section → one sentence establishing service area and response coverage.

Keep the existing conversational voice. It is a genuine differentiator; the competitors write vendor boilerplate.

8.4 Video SEO (significant, currently unexploited)

Five autoplaying videos, each with `aria-label` but **no structured data and no transcript**. `ServiceVideo` does not emit `VideoObject` markup.

Recommended:

1. Wrap each in `VideoObject` JSON-LD (`name`, `description`, `thumbnailUrl`, `uploadDate`, `duration`, `contentUrl`).
2. Add `public/videos/<service>/transcript.txt` and link it as the text alternative.

3. Keep `preload="metadata"` — already correct for bandwidth.

Real user impact is secondary here — no customer will search for a transcript. The value is entity richness and LLM citability.

8.5 Performance — the video is the LCP risk

The services page stacks up to five autoplaying videos totalling ~12 MB, several simultaneously in view. This directly threatens Core Web Vitals, which are a ranking input.

- Confirm each video is lazy-started — only play when in view (`preload="metadata"` , but `autoplay` is unconditional).
- Serve WebM/AV1 with MP4 fallback via `<source>` .
- Poster images (`poster` attribute) so a paused/unsupported video is not an empty box.
- Target LCP < 2.5s, CLS < 0.1, INP < 200ms; measure with §10 tooling.

8.6 Internal linking

- Home page bento cards → the matching `/services/*` page (currently the bento cards are **not** links at all; verified in `features-section-demo-3.tsx`).
- Footer "SERVICES" column → `/services` plus each child page.
- `service-section.tsx` → its own child page via a "All CCTV services →" link.

9. Structured data ●

Zero JSON-LD currently. Add in priority order.

9.1 ProfessionalService — every page, via a shared component

Highest-value single change. Makes the business entity explicit for local results and AI surfaces.

```
// components/json-ld.tsx
export function OrganizationSchema() {
  const data = {
    "@context": "https://schema.org",
    "@type": "ProfessionalService",
    "@id": "https://nanocomputingict.com/#organization",
    name: "Nano Computing ICT Solutions",
    url: "https://nanocomputingict.com",
    logo: "https://nanocomputingict.com/nano-logo.png",
    image: "https://nanocomputingict.com/nano-logo.png",
    description:
      "CCTV and security camera installation, door access control, time and attendance systems, computer network design and installation, server infrastructure, web and mobile development in Addis Ababa, Ethiopia.",
    telephone: "+251923787878",
    email: "info@nanocomputingict.com",
    address: {
      "@type": "PostalAddress",
      streetAddress: "Rayuma Building, Airport Road",
      addressLocality: "Addis Ababa",
      addressCountry: "ET",
    },
    areaServed: [
      { "@type": "City", name: "Addis Ababa" },
      { "@type": "Country", name: "Ethiopia" },
    ],
    knowsAbout: [
      "CCTV installation", "Security camera systems", "Door access control",
      "Biometric access control", "Time and attendance systems",
      "Structured network cabling", "Server infrastructure",
      "Web development", "Mobile app development", "Computer repair",
    ],
    openingHours: "Mo-Sa 08:00-18:00",
  };
  return (
    <script
      type="application/ld+json"
      dangerouslySetInnerHTML={{ __html: JSON.stringify(data) }}
    />
  );
}
```

Render once in `app/layout.tsx` .

Confirm all NAP data before publishing. Phone, address and hours above are taken from `app/layout.tsx` and `contact-section.tsx` in this codebase. `ProfessionalService` is the most sensitive schema type — inconsistent NAP across schema, Google Business Profile and the website is the classic cause of local ranking loss. Verify every field against the GBP listing before shipping. If the GBP address differs from `Rayuma Building, Airport Road` , resolve that first.

9.2 BreadcrumbList

On every non-home route. Cheap, and improves SERP presentation.

9.3 Service + VideoObject

On `/services` and each `/services/*` page (§8.2, §8.4).

9.4 FAQPage

Only if the visible FAQ content genuinely exists. Marking up content that isn't on the page risks a manual action — do not fabricate.

9.5 Validation

```
curl -X POST https://validator.schema.org/validate \
-d "output=json" \
--data-urlencode "url=https://nanocomputingict.com/services"
```

Verified: this endpoint works with **no API key** and `isRendered: true` , so it catches schema injected by client JS. Validate every URL type before shipping.

10. Tooling stack

Researched and verified 2026-10-03. Free tiers listed first.

10.1 Agent skills (procedural knowledge — free)

Install: `npx skills add <owner/repo> --skill <name>`

SKILL	INSTALLS	USE
seo-audit	219K	5-layer audit: crawlability, technical, on-page, content, authority
programmatic-seo	139K	Template page generation — relevant at §8.2 scale
ai-seo	135K	AI Overviews / ChatGPT / Perplexity citation optimisation
seo	48K	Evidence-led: runs live Lighthouse, then checks signals Lighthouse can't
seo-local	6.0K	Local SEO — most relevant skill for this business
google-search-console	1.8K	Reading GSC reports

Most valuable here: `coreyhaines31/marketingskills` (52,473★, MIT, pushed same day) and `AgriciDaniel/claude-seo` (18,192★, 20+ granular skills).

Skills are **procedural, not data**. They cannot supply search volume or competitor metrics. They tell an agent *how* to audit; §10.2–10.3 supply the *what*.

10.2 MCP servers

SERVER	COST	NOTES
Google Search Console via lionkiii/google-searchconsole-mcp	Free	Easiest auth — browser OAuth, no Google Cloud project
mikushnuz/gsc-mcp	Free	13 tools, broadest coverage (URL inspection, sitemaps)
ahonn/mcp-server-gsc	Free	Most popular (275★)
DataForSEO — https://mcp.dataforseo.com/v3/mcp	\$50 min top-up	Live SERP + volume + difficulty at ~\$0.002/query. The one paid recommendation
GA4 — googleanalytics/google-analytics-mcp	Free	Official Google server, 3,363★. Read-only, experimental

There is no official Google Search Console MCP server. Google ships 50+ MCP servers (github.com/google/mcp) and Search Console is not among them — verified independently.

Do not use [egebese/dataseo-mcp](#). It scrapes Ahrefs' free interface via CAPTCHA-solving services and disclaims ToS compliance. Never on client work.

10.3 Free keyword research — verified working, no API key

METHOD	ENDPOINT	VERIFIED
Google Suggest	suggestqueries.google.com/complete/search?client=chrome&hl=en&gl=et&q=	854 live calls
Geo-subtype capture	same, client=firefox	Returns google:suggestsubtypes — Google's own geo flag
Wikipedia opensearch	en.wikipedia.org/w/api.php?action=opensearch	Descriptive UA required or it throttles
Google Trends	trendspyg — https://github.com/flack0x/trendspyg	Free MCP, no key

Dead ends — do not build on these:

- **Google Programmable Search JSON API is closed to new customers** (official docs, updated 2026-02-18). Existing customers must migrate by 2027-01-01.
- **pytrends is archived.** Google rotated its cookies. Use [trendspyg](#).
- **Brave Search API free tier eliminated Feb 2026** — \$5/1,000 req, card required.
- **Google Keyword Planner** has no programmatic volume API for non-advertisers.

10.4 Technical auditing

TOOL	COST	USE
unlighthouse	Free	Crawls the whole site with Lighthouse. No key, no quota — the PSI escape hatch
@lhci/cli	Free	Per-commit Lighthouse in CI. Stops regressions
validator.schema.org	Free	POST /validate , renders JS, no key . Primary structured-data check
PageSpeed Insights API	Free with key	Without a key: HTTP 429 , tested. With a key: 25,000/day
CrUX API	Free with key	Real-user CWV. 150 queries/min, no daily cap
Sitebulb	14-day trial	MCP is read-only — cannot start crawls

Screaming Frog has no free CLI. Use Unlighthouse + LHCI. `audit-ci` is the *accessibility* CI action, not SEO. Easy to confuse with LHCI.

10.5 Google Search Console quotas

Source: <https://developers.google.com/webmaster-tools/limits>

RESOURCE	LIMIT
Search Analytics	1,200 QPM per site/user
URL Inspection	600 QPM, 2,000 QPD per site
Row cap	25,000 rows per request
Daily row cap	~50,000 rows/day/search type/property
Data retention	~16 months
Reporting lag	2–3 days

The row cap is why GSC totals rarely reconcile with the dashboard. For a small site the QPM limits are irrelevant; **the row cap and the 2–3 day lag are the real constraints** — design dashboards around weekly, not daily, movement.

11. Local SEO ●

This is a local business and local ranking is the whole game. `ProfessionalService` schema (§9.1) is necessary but not sufficient — it must agree with Google Business Profile.

1. **Claim and complete GBP.** Nano appears to hold a **5.0 rating** — real and valuable, but invisible without a complete profile.
2. **Exact NAP parity** across GBP, schema, footer and every page. Inconsistency is the leading cause of local ranking loss.
3. **Category selection.** Primary "Security system installer" / "Computer support and services"; secondary "Electrician" or "Telecommunications service" where genuinely applicable.
4. **Geo property:** use a **Domain property** (`sc-domain:nanocomputingict.com`) so www/non-www share one credential and one Sitemap.

5. **Photos:** the shop interior (public/nano-1.jpg), installed CCTV, racks. Real local photos outperform stock.
6. **Reviews:** ask after job completion. A 5.0 with volume beats 4.9 without.

12. Prioritised roadmap

P0 — Unblock indexing (do first, ~2 hours)

#	ACTION	FILE
1	Remove root alternates.canonical ●	app/layout.tsx:68-70
2	Set metadataBase	app/layout.tsx
3	Replace the rotating h1 with a static keyword h1 ●	components/home-hero.tsx:35-44
4	Verify each route emits a self-referencing canonical	—

Without #1 nothing else can be measured.

P1 — Foundation (~1 day)

1. app/sitemap.ts · 6. app/robots.ts · 7. app/opengraph-image.tsx · 8. ProfessionalService + BreadcrumbList JSON-LD · 9. Custom app/not-found.tsx · 10. Confirm live NAP matches schema before shipping #8.

P2 — Content architecture (~1 week)

1. Build the six /services/* pages (§8.2) · 12. Add location language to /services copy (§8.3) · 13. Make bento cards link to their service pages (§8.6) · 14. Build out or noindex /about and /shop (§5) · 15. Add /pricing targeting the price in ethiopia cluster (§6.3).

P3 — Enrichment (~1 week)

1. VideoObject markup + transcripts (§8.4) · 17. FAQ blocks with matching FAQPage markup · 18. GBP optimisation and NAP audit (§11) · 19. Link building — no local competitor has published pricing.

P4 — Measurement (ongoing)

1. Connect GSC via MCP (§10.2) · 21. CrUX + LHCI baselines (§10.4) · 22. Weekly GSC review; expand via Suggest as new pages ship (§10.3).

Expected sequencing

P0–P1 is under a day and fixes every structural defect. P2 is where ranking movement becomes possible. **Do not measure before P0 lands** — a canonical bug makes post-fix improvement impossible to attribute.

13. Verification checklist

Run after every change.

```
# 1. Per-route canonical must be self-referencing and unique
for r in / /services /contacts /about /shop; do
  echo -n "$r → "
  curl -s "https://nanocomputingict.com$r" \
    | grep -o '<link rel="canonical" href="[^"]*"'
done
# Expect: /services → .../services (NOT the homepage)

# 2. No absolute localhost URLs in metadata
curl -s https://nanocomputingict.com/services | grep -c 'localhost' # expect 0

# 3. Sitemap and robots resolve
curl -sI https://nanocomputingict.com/sitemap.xml | head -1 # 200
curl -s https://nanocomputingict.com/robots.txt

# 4. Structured data valid (no key)
curl -s -X POST https://validator.schema.org/validate \
  -d "output=json" --data-urlencode "url=https://nanocomputingict.com/services"

# 5. Exactly one h1 per page, and it is static
curl -s https://nanocomputingict.com/ | grep -c '<h1' # expect 1
```

Plus, in CI:

```
npx unighthouse --site https://nanocomputingict.com
npx @lhci/cli autorun
```

14. Honest limitations

Stated plainly, because a plan that overstates its evidence is worse than no plan:

1. **No search volume data exists anywhere in this document.** Google Suggest provides none. Every demand signal here is *suggest frequency across seeds* — a proxy. **Real volumes require DataForSEO (~\$50) or Keyword Planner.** Prioritise accordingly: validate the §6.2 "zero-suggestion" opportunities, since those rest on the strongest inference (no competition) rather than on volume.
2. **No keyword difficulty data was obtained.** "Low competition" for the zero-suggestion terms is inferred from absence of competitors, not measured.
3. **Local ranking is modelled on documented Google behaviour, not on this site's data.** No GSC property is connected, so there is no baseline. P4 exists to establish one.
4. **Traffic projections are deliberately absent.** Without volume and difficulty data any forecast would be invention.
5. **The live WordPress site has known defects** found during competitor research — placeholder theme text ("This is the heading / Offering the best image box wordpress plugin") on the homepage, a mislabeled Computer Repair slider, broken `about.html` links. Any migration should not carry these forward.
6. **§9.1's NAP values are taken from this codebase and are unverified against Google Business Profile.** Publishing inconsistent NAP is actively harmful. Confirm first.
7. **Competitor findings are from their public marketing copy and page structure,** not from backlink or traffic data — neither of which was obtainable for free.

15. Sources

Verified APIs — Google Suggest (suggestqueries.google.com , 854 calls) · Google Suggest firefox client (132 calls, geo subtypes) · Wikipedia opensearch · validator.schema.org/validate · PageSpeed Insights API (no-key quota test) · GSC quotas doc · SEO MCP endpoints (live [initialize](#) probes, all returned 401 as expected for unauthenticated)

Verified repos — checked via GitHub API for stars/licence/last-push: [coreyhaines31/marketingskills](#) 52,473★ · [AgriciDaniel/claude-seo](#) 18,192★ · [every-app/open-seo](#) 22,200★ · [addyosmani/web-quality-skills](#) 2,867★ · [googleanalytics/google-analytics-mcp](#) 3,363★ · [ahonn/mcp-server-gsc](#) 275★ · [harlan-zw/unlighthouse](#) 4,878★ · [GoogleChrome/lighthouse-ci](#) 7,102★ · [pa11y](#) 4,568★ · [dequelabs/axe-core](#) 7,587★ · [flack0x/trendspyg](#) 53★

Explicitly NOT verified — Bing autosuggest behaviour · [Autom8Minds/seo-mcp](#) capability claims (2★) · [loalseodata/mcp-server](#) uptime and pricing · Sitebulb headless/CLI mode · Google Keyword Planner programmatic access · Semrush/Ahrefs plan tier claims for MCP access (secondary source)

Do not use — [egebese/dataseo-mcp](#) (CAPTCHA-solving against Ahrefs ToS) · Google Programmable Search JSON API (closed to new customers) · [pytrends](#) (archived) · any agent-marketed "SEO tool" repo under 10★